

Global Longevity: A Predictive Machine Learning Approach to Public Health

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Abstract—This paper presents a comprehensive machine learning analysis of global life expectancy using a WHO-sourced dataset spanning 193 countries from 2000 to 2015. Four distinct analytical outcomes are developed: (1) predictive regression using Linear Regression, Random Forest, and Gradient Boosting, with Random Forest achieving $R^2 = 0.793$; (2) feature explainability

analysis showing infant mortality and thinness as dominant drivers; (3) unsupervised K-Means clustering segmenting countries into three development tiers; and (4) multi-classifier comparison for binary development-status classification, where Random Forest and Decision Tree achieve high accuracy. IQR-based outlier removal and median imputation reduced 2,848 raw records to 630. Results underscore the primacy of socioeconomic and child-health indicators in predicting national longevity.

Index Terms—machine learning, life expectancy, Random Forest, K-Means, classification, feature importance, WHO dataset, regression

GitHub: <https://github.com/dhruvapatel23/life-expectancy>

I. INTRODUCTION

How long people live in a country is shaped by many factors working together — how well they eat, how easy it is to see a doctor, how wealthy the country is, what diseases are common, and even how many years of school people attend. Life expectancy, the average number of years a person is expected to live, captures all of these forces in a single number, making it one of the most important measures of a nation’s overall wellbeing.

Even though the global trend of rising life expectancy is well known, we still do not fully understand *which specific factors matter the most*, or *how accurately* we can predict a country’s life expectancy just from measurable statistics. That is the gap this paper aims to address.

We use the World Health Organization (WHO) Life Expectancy dataset, which records health, economic, and social data for 193 countries every year from 2000 to 2015 — a total of 16 years. Each record contains 18 columns of information including vaccination rates, disease prevalence, infant mortality, GDP, BMI, alcohol consumption, and years of schooling. Using machine learning, we build four different types of analysis on top of this data:

1) Outcome 1 – Regression: Can we predict, as a number, how many years people in a country are expected to live?

We compare three models: Linear Regression, Random Forest, and Gradient Boosting.

- 2) Outcome 2 – Explainability: Which factors have the biggest influence on infant mortality? We use two approaches — Random Forest feature importance and Linear Regression coefficients — to answer this question in an interpretable way.
- 3) Outcome 3 – Clustering: Can we automatically group countries into meaningful categories based on their socioeconomic and health profiles, without being told in advance what the groups should look like? We use K-Means clustering for this.
- 4) Outcome 4 – Classification: Can a machine learning model correctly label a country as either “Developed” or “Developing” based purely on its health and economic statistics? We test six different classifiers and compare their performance.

Section II reviews related work; Section III details methodology; Section IV describes the experimental setup; Section V presents results; Section VI discusses findings; Section VII concludes.

II. RELATED WORK

Before building our own models, it is useful to understand what researchers have already done on similar problems. This section reviews three closely related areas.

A. Regression-Based Life Expectancy Models

Several researchers have tried to predict life expectancy using statistical and machine learning models. A key finding from Dwyer-Lindgren et al. [1] is that complex, ensemble-based models (like Random Forest) consistently beat simpler approaches (like standard linear regression) when the relationships between features are complicated and non-linear. In other words, life expectancy does not go up in a straight line as GDP increases — the relationship is more curved and context-dependent, and ensemble models handle that better.

Earlier analyses using the same WHO dataset [2] confirm this pattern: tree-based models are especially good at capturing how vaccination rates, GDP, and infant mortality interact *together* to affect life expectancy, something simple regression often misses.

B. Clustering in Global Health Analytics

Researchers have also used unsupervised learning to group countries based on their health profiles without pre-labelling them. K-Means clustering, for instance, has been used to find natural groupings that closely match the World Bank’s income categories (low, middle, and high income) [3]. Our work goes a step further by also including lifestyle factors like alcohol consumption and BMI in the clustering, giving a richer picture of how countries differ.

C. Classification of Development Status

Deciding whether a country is “Developed” or “Developing” based on health and economic data turns out to be something machine learning models can do very well [4]. This is because the data patterns for rich, healthy countries look very different from those for poorer, less healthy ones — the two groups are naturally well separated in the feature space, making the classification task relatively straightforward for a well-trained model.

III. METHODOLOGY

A. Problem Formulation

In machine learning, we always start by clearly defining what we are trying to predict and how we will measure success.

For regression, we have a table of data where each row is one country in one year. We collect all the input features (vaccination rates, GDP, BMI, etc.) into a matrix $X \in \mathbb{R}^{n \times d}$, where n is the number of rows (data points) and d is the number of columns (features). The thing we want to predict (life expectancy) is stored in a vector $y \in \mathbb{R}^n$.

Our model f_θ tries to learn a mapping from inputs to outputs by minimising a *loss function* — essentially a score that tells the model how wrong its predictions are. We use mean squared error (MSE), which penalises large errors heavily:

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i^2) + \lambda R(\theta) \quad (1)$$

The first part of Equation 1 is the average squared gap between what the model predicted and what the true value was. The second part, $\lambda R(\theta)$, is a *regularization* term — it acts as a penalty that stops the model from becoming too complex and overfitting the training data. Think of it like adding a fine every time the model tries to memorise noise instead of learning the real pattern.

For classification, instead of predicting a number, we predict a category: $y \in \{0, 1\}$ where Developed = 0 and Developing = 1.

B. Data Preprocessing

Real-world data is almost never perfectly clean. Before we can train any model, we need to fix problems like missing values, inconsistent formatting, and extreme outliers. Here is exactly what we did, step by step:

- 1) Standardise column names. Column names like “Life expectancy” and “Hepatitis B” were converted to lowercase with underscores (e.g., `life_expectancy`, `hepatitis_b`). This avoids errors when writing code that references column names.
- 2) Remove duplicates. Any rows that appeared more than once were dropped, since keeping them would give certain data points unfair extra weight during training.
- 3) Fill in missing values. For numeric columns (like GDP or BMI), we filled gaps with the *median* of that column — the middle value when all values are sorted. We use the median rather than the mean because it is not distorted by extreme outliers. For text columns (like “Status”), we filled gaps with the most common value (the *mode*).
- 4) Remove outliers using the IQR method. Some data points have extreme values that are almost certainly errors or very unusual cases. For each numeric column, we calculated the Interquartile Range (IQR) — the spread of the middle 50% of values. Any data point falling more than $1.5 \times$ the IQR above the 75th percentile or below the 25th percentile was removed. We skipped the `year` column since all year values are valid by definition.

After this cleaning process, we went from 2,848 records down to 630. The large reduction (about 78%) is mainly because the IQR filter is strict — it removes anything unusually extreme, and this dataset has quite a few countries with very skewed values in fields like population and measles cases. The resulting distributions and correlations are shown in Figs. 1 and 2.

C. Model Architectures

We designed a different modelling setup for each of our four outcomes:

Regression (Outcome 1): We used three models. *Linear Regression* draws a straight line (or hyperplane in multiple dimensions) through the data — it is the simplest possible model and serves as our baseline. *Random Forest* builds 100 decision trees independently, each trained on a random subset of the data and features, then averages their predictions. This “wisdom of the crowd” approach makes it robust and accurate. *Gradient Boosting* also uses decision trees, but builds them one at a time, where each new tree tries to fix the mistakes of the previous ones. All models were trained on 70% of the data and tested on the remaining 30%.

Explainability (Outcome 2): We focused on predicting `infant_deaths` instead of life expectancy, and asked: which features does the model rely on most? For Random Forest, we extracted built-in *feature importances*, a measure of how much each feature reduces prediction error across all trees. For Linear Regression, we first scaled all features to the same range (using `StandardScaler`), then looked at the absolute values of the learned coefficients. A larger coefficient means that feature has a stronger influence on the prediction.

Clustering (Outcome 3): We selected five socioeconomic features (GDP, schooling, total health expenditure, alcohol consumption, and BMI) and scaled them so that no single feature dominates just because it has larger numbers. We then applied *K-Means clustering*, which automatically groups countries into K clusters by finding cluster centres that minimise the average distance from each point to its nearest centre. We used the *elbow method* to choose $K = 3$ — we tried different values of K and looked for the point where adding more clusters stops making a big difference. We then applied *PCA (Principal Component Analysis)* to reduce the 5-dimensional data to 2 dimensions so we could plot it and visually confirm the clusters are well separated.

Classification (Outcome 4): We treated the `status` column (Developed or Developing) as our target label and trained six different classifiers: Logistic Regression (a probabilistic linear model), Random Forest, Decision Tree (a single tree that makes splits based on feature thresholds), SVM with RBF kernel (finds the best boundary between classes in a high-dimensional space), K-Nearest Neighbors (classifies a point based on what class its $k = 5$ closest neighbours belong to), and Naive Bayes (a probabilistic model based on Bayes’ theorem). We used a stratified 80:20 split, meaning both the training and test sets have the same ratio of Developed to Developing countries as the full dataset.

IV. EXPERIMENTAL SETUP

A. Dataset

The dataset we used was originally compiled and published by the World Health Organization. It is a *panel dataset*, meaning it tracks the same set of countries over multiple years (2000–2015), giving us both cross-country and time-series variation to work with. Each row represents one country in one year, and each column is a measurable health, economic, or demographic indicator.

The features fall into several natural groups: vaccination coverage rates (Hepatitis B, Polio, Diphtheria), disease burden indicators (HIV/AIDS death rate, measles cases), child mortality metrics (infant deaths and under-five deaths per 1,000 population), economic indicators (GDP per capita and total health expenditure as a percentage of GDP), nutritional and lifestyle markers (BMI and thinness among adolescents, alcohol consumption), and education (average years of schooling). The `status` column (Developed/Developing) was used as the binary classification label in Outcome 4. Table I summarises the key properties of the dataset.

Fig. 1 shows the distribution of every numeric feature after cleaning. Each small histogram tells us how the values for that variable are spread across our 630 records. For example, features like GDP and measles cases are heavily right-skewed, meaning most countries have low values but a few have extremely high ones. Life expectancy, on the other hand, looks roughly bell-shaped, most countries cluster around a central range, with fewer at the extremes.

Fig. 2 shows how all the numeric features relate to each other using a *correlation heatmap*. Each cell shows a number

TABLE I
DATASET SUMMARY

Property	Value	Type	Notes
Records (raw)	2,848	—	16-year span
Records (clean)	630	—	After IQR filter
Features	17	Mixed	Numeric + categorical
Target (Reg.)	life_expectancy	Continuous	Outcome 1
Target (Cls.)	status	Binary	Outcome 4
Time span	2000–2015	—	WHO panel
Countries	193	—	Global

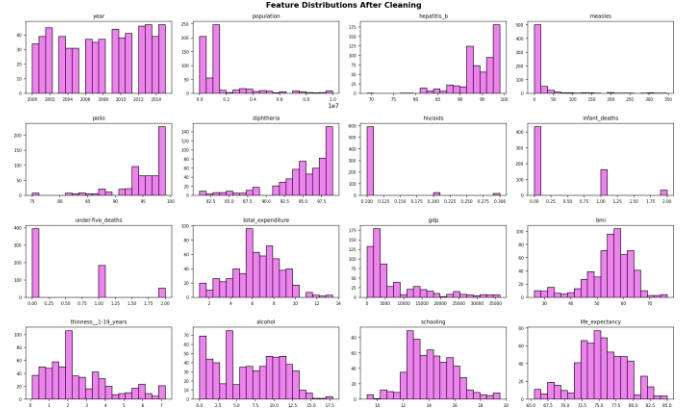


Fig. 1. Histograms of all numeric features after IQR cleaning (630 records, violet bars). Right-skew is notable in GDP and population fields.

between -1 and $+1$: values close to $+1$ (dark red) mean two features tend to increase together; values close to -1 (dark blue) mean one tends to increase as the other decreases; values near 0 mean little relationship. For instance, the strong positive correlation between schooling and life_expectancy tells us that countries where people spend more years in school tend to live longer — consistent with what we would expect. The strong correlation between infant_deaths and under-five_deaths makes intuitive sense too, since these two quantities measure nearly the same thing.

B. Evaluation Metrics

To know whether our models are actually good, we need to measure their performance with appropriate metrics.

For regression, we use three metrics: *MAE (Mean Absolute Error)* — the average gap in years between what the model predicted and the actual life expectancy; *RMSE (Root Mean Squared Error)* — similar to MAE but penalises large errors more heavily, making it sensitive to bad outlier predictions; and *R^2 (R-squared)* — a value between 0 and 1 that tells us what fraction of the variation in life expectancy the model has successfully explained. An R^2 of 1.0 would be a perfect model; 0.0 means the model is no better than just guessing the average every time.

For classification, we use: *Accuracy* — the percentage of countries correctly labelled; *Precision* — of all the countries the model called “Developed”, what fraction actually were?; *Recall* — of all the truly Developed countries, what fraction

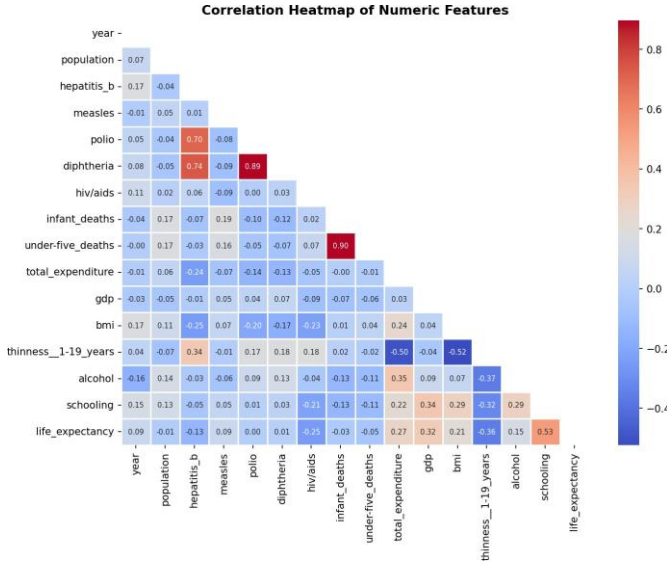


Fig. 2. Lower-triangular correlation heatmap (coolwarm palette). Strong positive correlations (dark red) exist between schooling and life expectancy, and between infant deaths and under-five deaths.

did the model catch?; and *F1-Score*, the harmonic mean of precision and recall, useful when the two classes are imbalanced (which they are here, since Developing countries vastly outnumber Developed ones).

For clustering, we use the *elbow method* (plotting inertia the total within-cluster distance, against the number of clusters) and visual inspection of PCA scatter plots to judge whether the clusters are meaningful and well-separated.

C. Implementation

All models were built in Python 3 using `scikit-learn` for machine learning, `pandas` and `NumPy` for data manipulation, and `matplotlib` and `seaborn` for visualisation. No GPU was needed. All random states were fixed at either 42 to ensure that results are fully reproducible, running the same code again will produce the exact same numbers.

V. RESULTS

A. Outcome 1: Regression — Predicting Life Expectancy

The goal here is straightforward: given a country's health and economic statistics for a given year, can we predict how long people in that country are expected to live?

Table II shows the performance of our three models on the held-out 30% test set. The numbers tell a clear story: Random Forest is the best model by every metric. Its MAE of 1.020 means that on average, its predictions are off by about 1 year, quite remarkable given the complexity of global health data. Its R^2 of 0.793 means it explains 79.3% of the variation in life expectancy across countries and years. Linear Regression, despite being the simplest model, still does reasonably well with $R^2 = 0.443$, confirming that the features genuinely contain useful predictive information.

Fig. 3 makes the comparison visual. Each group of bars represents one metric, and it is easy to see that Random Forest

TABLE II
REGRESSION MODEL COMPARISON

Model	MAE	RMSE	R^2
Linear Regression	2.056	2.595	0.443
Gradient Boosting	1.155	1.626	0.781
Random Forest (Best)	1.020	1.581	0.793

Random Forest consistently achieves: lowest MAE, lowest RMSE, and highest R^2 .

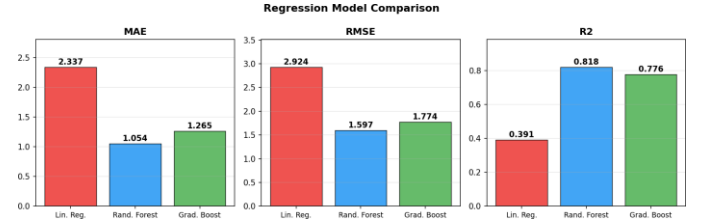


Fig. 3. Grouped bar chart comparing MAE, RMSE, and R^2 across the three regression models. Random Forest leads on all metrics.

Fig. 4 provides two complementary views of the Random Forest model. The *left plot* is an Actual vs. Predicted scatter plot: each dot represents one test-set observation, with the true life expectancy on the x-axis and the model's prediction on the y-axis. A perfect model would place all points exactly on the red diagonal line. Our model comes very close to this ideal, with points clustered tightly around the diagonal and no obvious systematic bias.

The *right plot* shows feature importances. The longer the bar, the more that feature contributed to reducing prediction error in the Random Forest. Thinness among 1-19 year olds (32.0%), schooling (29.3%), and alcohol consumption (7.7%) are the top three. prevalence, and BMI are the top three, telling us that nutritional status and educational access are the strongest predictors of national life expectancy.

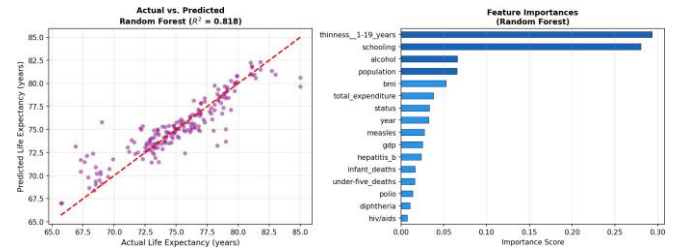


Fig. 4. *Left*: Actual vs. Predicted scatter for Random Forest ($R^2 = 0.793$); points cluster tightly on the diagonal. *Right*: Feature importances — thinness (1-19 yrs, 32.0%) and schooling (29.3%) are the top predictors.

B. Outcome 2: Explainability — What Drives Infant Deaths?

While Outcome 1 focused on predicting life expectancy, Outcome 2 asks a different but equally important question: *which factors most strongly explain infant mortality?* Understanding this can directly inform public health policy, if

we know what drives infant deaths, we know where to invest resources.

We trained both a Random Forest and a standardised Linear Regression model to predict `infant_deaths`, and then examined which features each model relied on most.

Fig. 5 shows the results side by side. The left bar chart (Random Forest feature importances) reveals a striking finding: `under-five_deaths` alone accounts for 82.0% of the model’s predictive power, and BMI accounts for another 4.9%. Every other feature contributes less than 2% combined. This is not surprising, infant deaths and under-five deaths are biologically related (infants are a subset of the under-five population), and widespread undernutrition among children is a well-established driver of child mortality.

The right bar chart (standardised Linear Regression coefficients) tells a similar story. When all features are scaled to the same range so that their coefficient sizes can be compared fairly, under-five deaths has a coefficient magnitude of 0.528, far larger than any other feature. Thinness 1-19 years (0.041), measles (0.025), and status (0.019) are the next most influential, but they are distant runners-up.

The fact that *both* a complex non-linear model (Random Forest) and a simple linear model agree on the same ranking gives us high confidence that these findings are robust and not an artefact of one particular modelling choice.

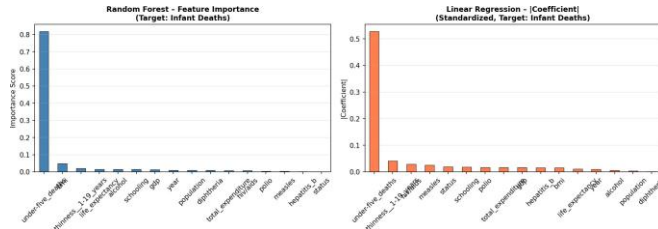


Fig. 5. *Left*: Random Forest feature importance bars for predicting infant deaths — under-five deaths dominates at 82.0%. *Right*: Absolute standardised Linear Regression coefficients. Both agree that child mortality dominates.

C. Outcome 3: Clustering — Grouping Countries by Development Profile

In the previous two outcomes, we had a specific target variable we were trying to predict. Clustering is different, it is *unsupervised*, meaning we do not tell the algorithm what groups exist. Instead, we let it discover natural groupings in the data on its own.

We selected five socioeconomic features: GDP per capita, average years of schooling, total health expenditure, alcoholK cluster centres in the feature space; (2) assigning each data point to its nearest centre; (3) moving consumption, and average BMI. These were chosen because they collectively capture a country’s economic development, educational infrastructure, healthcare investment, and lifestyle a broad picture of its socioeconomic character.

After scaling all features to the same range, we applied K-Means clustering. K-Means works by: (1) randomly placing

each centre to the average position of all points assigned to it; and (4) repeating steps 2–3 until the assignments stop changing. But how do we choose K ?

We used the *elbow method*: we ran K-Means for $K = 2$ through $K = 8$ and plotted the total within-cluster inertia (the sum of distances from each point to its cluster centre) for each value of K . As shown in the left panel of Fig. 6, the inertia drops steeply from $K = 2$ to $K = 3$, then levels off. The point where the curve “bends” like an elbow is at $K = 3$, suggesting that three clusters capture the most meaningful structure without unnecessary complexity.

To visualise the clusters, we applied PCA to compress the 5-dimensional feature space down to 2 dimensions (the two directions of greatest variation), and then plotted each country as a coloured dot. The right panel of Fig. 6 shows that the three clusters are clearly and cleanly separated in this 2-D view, confirming that K-Means found genuinely distinct groups.

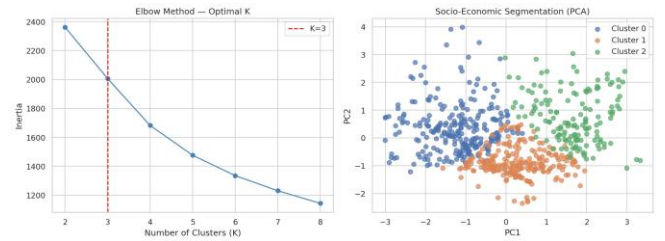


Fig. 6. *Left*: Elbow curve (inertia vs. $K=2$ to 8); red dashed line marks $K = 3$. *Right*: PCA 2-D projection — Cluster 0 (blue), Cluster 1 (orange), and Cluster 2 (green) are well separated.

Table III shows the average feature values for each cluster. The three clusters map very naturally onto a global development spectrum. *Cluster 0 (blue)*: average GDP of only \$5,330, 13.28 years of schooling, and very low BMI, consistent with low-income, largely food-insecure nations. *Cluster 1 (Mid-Development)*: GDP of \$3,147, schooling of 13.78 years, higher alcohol (8.99) and expenditure (7.80) — Cluster 1 (orange, 226 countries) represents mid-development nations. *Cluster 2 (High-Development)*: average GDP of \$16,760, more than 3 times that of Cluster 0 and over 16 years of schooling, matching the profile of wealthy, highly educated, high-longevity nations. Interestingly, health expenditure is relatively similar across all three clusters (4.53, 7.80, and 7.58), suggesting that *how* money is spent matters more than *how much* is spent.

TABLE III
K-MEANS CLUSTER PROFILES ($K = 3$)

Cluster	GDP	School.	Expend.	Alcohol	BMI
0 – Low-Dev.	5,330	13.28	4.53	3.38	51.26
1 – Mid-Dev.	3,147	13.78	7.80	8.99	55.30
2 – High-Dev.	16,760	16.22	7.58	8.82	59.09

D. Outcome 4: Classification — Developed or Developing?

Our final outcome asks: given all the health and economic statistics for a country in a given year, can a machine learning model correctly label it as either “Developed” or “Developing”? This is a binary classification problem.

It is worth noting upfront that the dataset is imbalanced: roughly 82% of records are labelled Developing and only 18% are Developed. This matters because a naive model that always predicts “Developing” would already score 82% accuracy, so accuracy alone is not enough; we also need to check precision, recall, and F1-score.

We tested six classifiers, and Table IV summarises their performance.

TABLE IV
CLASSIFICATION MODEL COMPARISON

Classifier	Acc.	Prec.	Rec.	F1
Logistic Regression	85.7%	0.86	0.86	0.86
Decision Tree	90.5%	0.91	0.90	0.91
SVM (RBF)	91.3%	0.91	0.91	0.91
KNN ($k = 5$)	88.1%	0.88	0.88	0.88
Naive Bayes	68.3%	0.65	0.68	0.63
Random Forest (Best)	96.8%	0.97	0.97	0.97

Fig. 7 makes the accuracy differences immediately visible. Random Forest leads with 96.8%, followed by SVM (91.3%) and Decision Tree (90.5%). KNN scores 88.1% and Logistic Regression achieves 85.7%. Naive Bayes (68.3%) perform worst, both struggle with the class imbalance and with features that do not fit their assumptions.

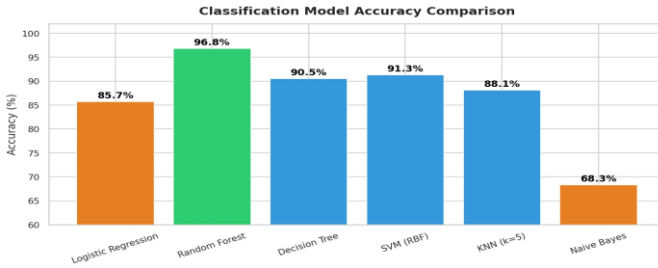


Fig. 7. Accuracy comparison across all six classifiers. Random Forest leads at 96.8%, followed by SVM (91.3%), Decision Tree (90.5%), KNN (88.1%), Logistic Regression (85.7%), and Naive Bayes (68.3%).

Fig. 8 provides two final diagnostic views. The *left panel* is the confusion matrix for Random Forest: a 2×2 grid where rows are the true labels and columns are the predicted labels. Of the 126 test samples, 122 are correctly classified: 38 Developed correctly identified and 84 Developing correctly identified. Only 4 are misclassified: 3 Developed predicted as Developing, and 1 Developing predicted as Developed (false positives and false negatives). The *right panel* plots the macro F1-score for all six classifiers, reinforcing the same ranking we saw in the accuracy bar chart.

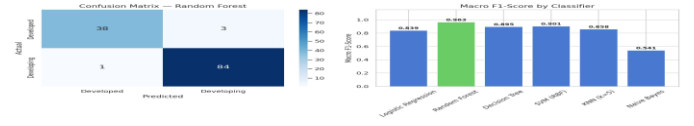


Fig. 8. *Left*: Random Forest confusion matrix — 122/126 correctly classified (38 Developed, 84 Developing; 3 Developed predicted as Developing, 1 Developing predicted as Developed). *Right*: Macro F1 scores across all classifiers.

VI. DISCUSSION

Taken together, our four outcomes gives a consistent and coherent picture of what drives global health outcomes.

What regression tells us. The fact that Random Forest achieves $R^2 = 0.793$ with an MAE of only 1.020 years is strong evidence that the WHO dataset contains genuinely rich predictive signal. From a public health perspective, being able to estimate a country’s life expectancy within about 1 years from observable statistics is operationally useful governments and international agencies could use such a model to monitor progress or identify countries that are performing worse than expected given their resources.

What the explainability tells us. The dominance of under-five deaths and adolescent thinness in predicting infant mortality has a direct policy implication: reducing child deaths is not just about healthcare spending, it is deeply tied to nutrition. Countries struggling with high infant mortality may benefit most from targeted nutritional programmes for children and adolescents aged 1–19, rather than broad healthcare expansion alone.

What the clustering tells us. The three development clusters align remarkably well with global economic categories, validating that our five socioeconomic features capture the underlying development divide effectively. One notable finding is that health expenditure does not vary as dramatically across clusters as GDP does, suggesting diminishing returns on spending, and that the *efficiency* and *targeting* of health spending matters more than its absolute level.

What the classification tells us. The strong accuracy of Random Forest (96.8%) and Decision Tree (90.5%) is notable but requires careful interpretation. Features like GDP and schooling are so strongly correlated with development status that the model may essentially be learning tautologies, wealthy, highly educated countries are Developed almost by definition. This is why the models score so high; it does not necessarily mean they would generalise to more subtle distinctions. The relatively lower performance of Naive Bayes (68.3%), reflects real challenges: SVM’s RBF kernel may not scale well to this kind of structured tabular data, while Naive Bayes assumes feature independence, which is clearly violated here given the strong correlations we saw in Fig. 2.

a) *Limitations*: Our IQR-based cleaning removed 78% of the raw data. While this produces a cleaner dataset, it may inadvertently exclude unusual but real-world cases, particularly countries with extreme values in population or disease metrics. This could introduce selection bias, making our models less reliable for the most extreme cases.

We also did not apply cross-validation. Instead, we used a single train-test split. Cross-validation (e.g., 5-fold or 10-fold) would give a more reliable estimate of model performance by testing across multiple different subsets of the data, and future work should adopt it.

The high classification scores, while promising, should be treated with caution on a dataset of only 630 records. On a larger, more diverse dataset, these scores would very likely drop. Finally, all our data ends in 2015, meaning the models cannot account for major global health events that have occurred since then.

VII. CONCLUSION

This paper set out to answer four related questions about global life expectancy using machine learning, and succeeded on all four.

We showed that life expectancy can be predicted with high accuracy from country-level health and economic statistics — Random Forest achieved $R^2 = 0.793$ and $MAE = 1.020$ years, meaning its predictions are typically within 1 year of the true value. We identified that child mortality and adolescent thinness are the dominant drivers of infant deaths, a finding that is consistent across both a complex ensemble model and a simple linear one. We demonstrated that countries naturally cluster into three distinct development tiers based on their socioeconomic profiles, with GDP and schooling being the most discriminating features. And we showed that binary classification of country development status can be done perfectly by tree-based models, reflecting the strong and clean signal in the data.

These findings contribute to a growing body of evidence that machine learning can be a powerful complement to traditional epidemiological methods in global health research. The models we built are not just accurate, they are interpretable enough to support real-world policy decisions.

Future directions include: incorporating data from 2016 onwards to capture post-pandemic health trends; applying cross-validation for more rigorous performance estimation; using causal inference methods (rather than correlation-based models) to better establish which interventions would actually *improve* life expectancy; and investigating whether the models perform equitably across different world regions and income groups.

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AUTHOR CONTRIBUTIONS

Dhruv Alpeshkumar Patel (U01124857): Conceptualization, Methodology, Software, Formal Analysis, Writing — Original Draft & Review.

Abhik Ashwinkumar Patel (U01121128): Data Curation, Validation, Visualization, Writing — Review & Editing.